PART 1 of 12

CLASS 10 SCIENCE

CHEMICAL REACTIONS AND EQUATIONS

MCQs 1-8

TOP 100 MCQs WITH ANSWERS

1. A chemical reaction is a process in which:

A) Only physical properties change
B) New substances are formed
C) Shape changes only
D) State changes only

Answer: B) New substances are formed

2. Which of the following indicates a chemical reaction?

A) Change in colour
B) Evolution of gas
C) Formation of precipitate
D) All of these

Answer: D) All of these

3. Magnesium ribbon burns in air to form:

A) Magnesium chloride
B) Magnesium sulphate
C) Magnesium oxide
D) Magnesium nitrate

Answer: C) Magnesium oxide

4. The chemical equation for burning magnesium is:

A) $Mg + O \rightarrow MgO$
B) $2Mg + O_2 \rightarrow 2MgO$
C) $Mg + O_2 \rightarrow MgO_2$
D) $MgO + O_2 \rightarrow Mg$

Answer: B) $2Mg + O_2 \rightarrow 2MgO$

5. Which law is satisfied by balancing a chemical equation?

A) Newton's Law
B) Law of Gravitation
C) Law of Conservation of Mass
D) Ohm's Law

Answer: C) Law of Conservation of Mass

6. The reactants in a chemical reaction are written:

A) On the right side
B) On both sides
C) On the left side
D) Below the arrow

Answer: C) On the left side

7. The products in a chemical equation are written:

A) On the right side
B) On the left side
C) Above the equation
D) Below the equation

Answer: A) On the right side

8. Which of the following is a balanced equation?

A) $H_2 + O_2 \rightarrow H_2O$
B) $H_2 + O_2 \rightarrow 2H_2O$
C) $2H_2 + O_2 \rightarrow 2H_2O$
D) $H_2 + 2O_2 \rightarrow H_2O$

Answer: C) $2H_2 + O_2 \rightarrow 2H_2O$

ANSWER KEY (PART 1)

Q. No.	1	2	3	4	5	6	7	8
Answer	B	D	C	B	C	C	A	C

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PART 2 of 12

CLASS 10 SCIENCE

CHEMICAL REACTIONS AND EQUATIONS

MCQs 9–16

TOP 100 MCQs WITH ANSWERS

9. Which of the following is a combination reaction?

A) $CaO \rightarrow Ca + O_2$
B) $2H_2 + O_2 \rightarrow 2H_2O$
C) $Zn + 2HCl \rightarrow ZnCl_2 + H_2$
D) $2KClO_3 \rightarrow 2KCl + 3O_2$

Answer: B) $2H_2 + O_2 \rightarrow 2H_2O$

10. Which of the following is a decomposition reaction?

A) $NaOH + HCl \rightarrow NaCl + H_2O$
B) $CaCO_3 \rightarrow CaO + CO_2$
C) $2Na + Cl_2 \rightarrow 2NaCl$
D) $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$

Answer: B) $CaCO_3 \rightarrow CaO + CO_2$

11. Which of the following is a displacement reaction?

A) $2H_2 + O_2 \rightarrow 2H_2O$
B) $Cu + 2AgNO_3 \rightarrow Cu(NO_3)_2 + 2Ag$
C) $2KClO_3 \rightarrow 2KCl + 3O_2$
D) $CaO + H_2O \rightarrow Ca(OH)_2$

Answer: B) $Cu + 2AgNO_3 \rightarrow Cu(NO_3)_2 + 2Ag$

12. Which of the following is a double displacement reaction?

A) $Zn + 2HCl \rightarrow ZnCl_2 + H_2$
B) $AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$
C) $2H_2 + O_2 \rightarrow 2H_2O$
D) $CaCO_3 \rightarrow CaO + CO_2$

Answer: B) $AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$

13. Which of the following is a redox reaction?

A) $NaCl + AgNO_3 \rightarrow AgCl + NaNO_3$
B) $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$
C) $CaCO_3 \rightarrow CaO + CO_2$
D) $HCl + NaOH \rightarrow NaCl + H_2O$

Answer: B) $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$

14. What happens when iron reacts with copper sulphate solution?

A) No reaction occurs
B) Iron sulphate and copper are formed
C) Iron oxide and copper are formed
D) Copper sulphate and iron are formed

Answer: B) Iron sulphate and copper are formed

15. Which gas is produced when zinc reacts with dilute HCl?

A) Oxygen B) Carbon dioxide
C) Hydrogen D) Nitrogen

Answer: C) Hydrogen

16. Balance the following equation: $Fe + H_2O \rightarrow Fe_3O_4 + H_2$

A) $3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$
B) $3Fe + 2H_2O \rightarrow Fe_3O_4 + 2H_2$
C) $Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$
D) $2Fe + 3H_2O \rightarrow Fe_3O_4 + 3H_2$

Answer: A) $3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$



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ANSWER KEY (PART 2)

Q.No.	9	10	11	12	13	14	15	16
Answer	B	B	B	B	B	B	C	A

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PART 3 of 12

CLASS 10 SCIENCE

CHEMICAL REACTIONS AND EQUATIONS

MCQs 17-25

TOP 100 MCQs WITH ANSWERS

17. Which of the following is an exothermic reaction?

A) Photosynthesis
B) Thermal decomposition of $CaCO_3$
C) Burning of coal
D) Electrolysis of water

Answer: C) Burning of coal

18. Which of the following is an endothermic reaction?

A) Respiration
B) Digestion of food
C) Formation of slaked lime
D) Thermal decomposition of $CaCO_3$

Answer: D) Thermal decomposition of $CaCO_3$

19. Which of the following is a combination reaction?

A) $CaO + CO_2 \rightarrow CaCO_3$
B) $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$
C) $2Na + 2H_2O \rightarrow 2NaOH + H_2$
D) $2KClO_3 \rightarrow 2KCl + 3O_2$

Answer: A) $CaO + CO_2 \rightarrow CaCO_3$

20. Which of the following is a decomposition reaction?

A) $2H_2 + O_2 \rightarrow 2H_2O$
B) $2H_2O_2 \rightarrow 2H_2O + O_2$
C) $Zn + 2HCl \rightarrow ZnCl_2 + H_2$
D) $CaO + H_2O \rightarrow Ca(OH)_2$

Answer: B) $2H_2O_2 \rightarrow 2H_2O + O_2$

21. Which of the following is a single displacement reaction?

A) $AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$
B) $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$
C) $2H_2 + O_2 \rightarrow 2H_2O$
D) $CaCO_3 \rightarrow CaO + CO_2$

Answer: B) $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$

22. Which of the following is a double displacement reaction?

A) $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$
B) $Na_2SO_4 + BaCl_2 \rightarrow BaSO_4 + 2NaCl$
C) $2KClO_3 \rightarrow 2KCl + 3O_2$
D) $Mg + 2HCl \rightarrow MgCl_2 + H_2$

Answer: B) $Na_2SO_4 + BaCl_2 \rightarrow BaSO_4 + 2NaCl$

23. Which of the following is a precipitation reaction?

A) $Na_2SO_4 + BaCl_2 \rightarrow BaSO_4 + 2NaCl$
B) $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$
C) $2H_2 + O_2 \rightarrow 2H_2O$
D) $2KClO_3 \rightarrow 2KCl + 3O_2$

Answer: A) $Na_2SO_4 + BaCl_2 \rightarrow BaSO_4 + 2NaCl$

24. Which of the following is a neutralization reaction?

A) $HCl + NaOH \rightarrow NaCl + H_2O$
B) $CaCO_3 \rightarrow CaO + CO_2$
C) $2H_2 + O_2 \rightarrow 2H_2O$
D) $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$

Answer: A) $HCl + NaOH \rightarrow NaCl + H_2O$

25. Identify the correct balanced equation for the reaction between iron and steam.

A) $Fe + H_2O \rightarrow Fe_3O_4 + H_2$
B) $3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$
C) $Fe + 2H_2O \rightarrow FeO + 2H_2$
D) $2Fe + 3H_2O \rightarrow Fe_2O_3 + 3H_2$

Answer: B) $3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$

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ANSWER KEY (PART 3)									
Q. No.	17	18	19	20	21	22	23	24	25
Answer	C	D	A	B	B	B	A	A	B

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PART 4 of 12

CLASS 10 SCIENCE

CHEMICAL REACTIONS AND EQUATIONS

MCQs 26-33

TOP 100 MCQs WITH ANSWERS

26. Which of the following is a single displacement reaction?

A) $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$
B) $2Na + Cl_2 \rightarrow 2NaCl$
C) $CaCO_3 \rightarrow CaO + CO_2$
D) $2AgNO_3 + Cu \rightarrow Cu(NO_3)_2 + 2Ag$

Answer: A) $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$

27. Which of the following is not a decomposition reaction?

A) $2H_2O_2 \rightarrow 2H_2O + O_2$
B) $CaCO_3 \rightarrow CaO + CO_2$
C) $2KClO_3 \rightarrow 2KCl + 3O_2$
D) $H_2 + Cl_2 \rightarrow 2HCl$

Answer: D) $H_2 + Cl_2 \rightarrow 2HCl$

28. Which of the following is an example of a combination reaction?

A) $C + O_2 \rightarrow CO_2$
B) $2HCl \rightarrow H_2 + Cl_2$
C) $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$
D) $CaO + H_2O \rightarrow Ca(OH)_2$

Answer: A) $C + O_2 \rightarrow CO_2$

29. Which of the following is an example of a redox reaction?

A) $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$
B) $HCl + NaOH \rightarrow NaCl + H_2O$
C) $CaCO_3 \rightarrow CaO + CO_2$
D) $2KClO_3 \rightarrow 2KCl + 3O_2$

Answer: A) $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$

30. Which of the following is a precipitation reaction?

A) $AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$
B) $HCl + NaOH \rightarrow NaCl + H_2O$
C) $2H_2 + O_2 \rightarrow 2H_2O$
D) $Zn + 2HCl \rightarrow ZnCl_2 + H_2$

Answer: A) $AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$

31. What happens when dilute HCl reacts with zinc granules?

A) No reaction occurs
B) Hydrogen gas is evolved
C) Oxygen gas is evolved
D) Carbon dioxide gas is evolved

Answer: B) Hydrogen gas is evolved

32. What is the colour of copper sulphate solution after an iron nail is placed in it for some time?

A) Blue B) Colourless
C) Green D) Yellow

Answer: B) Colourless

33. Which gas is evolved when marble chips ($CaCO_3$) are treated with dilute HCl?

A) Hydrogen B) Oxygen
C) Carbon dioxide D) Nitrogen

Answer: C) Carbon dioxide

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ANSWER KEY (PART 4)								
Q. No.	26	27	28	29	30	31	32	33
Answer	A	D	A	A	A	B	B	C

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PART 5 of 12

CLASS 10 SCIENCE

CHEMICAL REACTIONS AND EQUATIONS

MCQs 34-41

TOP 100 MCQs WITH ANSWERS

34. Which of the following is the correct balanced equation for the reaction of sodium with water?

A) $Na + H_2O \rightarrow NaOH + H_2$
B) $2Na + 2H_2O \rightarrow 2NaOH + 2H_2$
C) $2Na + H_2O \rightarrow NaOH + H_2$
D) $Na + 2H_2O \rightarrow 2NaOH + H_2$

Answer: B) $2Na + 2H_2O \rightarrow 2NaOH + 2H_2$

35. What type of reaction is $CaCO_3 \xrightarrow{\Delta} CaO + CO_2$?

A) Combination reaction
B) Decomposition reaction
C) Displacement reaction
D) Double displacement reaction

Answer: B) Decomposition reaction

36. Which of the following is a redox reaction?

A) $BaCl_2 + Na_2SO_4 \rightarrow BaSO_4 + 2NaCl$
B) $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$
C) $HCl + NaOH \rightarrow NaCl + H_2O$
D) $2KClO_3 \rightarrow 2KCl + 3O_2$

Answer: B) $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$

37. Which of the following reactions is not a combination reaction?

A) $2Mg + O_2 \rightarrow 2MgO$
B) $CaO + H_2O \rightarrow Ca(OH)_2$
C) $2Na + Cl_2 \rightarrow 2NaCl$
D) $CaCO_3 \rightarrow CaO + CO_2$

Answer: D) $CaCO_3 \rightarrow CaO + CO_2$

38. Why do we balance a chemical equation?

A) To make it look neat
B) To satisfy the law of conservation of mass
C) To increase its speed
D) To change the state of reactants

Answer: B) To satisfy the law of conservation of mass

39. Which of the following is a displacement reaction?

A) $Zn + 2HCl \rightarrow ZnCl_2 + H_2$
B) $2Na + Cl_2 \rightarrow 2NaCl$
C) $CaCO_3 \rightarrow CaO + CO_2$
D) $2H_2 + O_2 \rightarrow 2H_2O$

Answer: A) $Zn + 2HCl \rightarrow ZnCl_2 + H_2$

40. Which gas turns lime water milky?

A) Oxygen
B) Carbon dioxide
C) Nitrogen
D) Hydrogen

Answer: B) Carbon dioxide

41. Which of the following represents a neutralization reaction?

A) $H_2SO_4 + 2NaOH \rightarrow Na_2SO_4 + 2H_2O$
B) $2KClO_3 \rightarrow 2KCl + 3O_2$
C) $2H_2 + O_2 \rightarrow 2H_2O$
D) $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$

Answer: A) $H_2SO_4 + 2NaOH \rightarrow Na_2SO_4 + 2H_2O$

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ANSWER KEY (PART 5)

Q. No.	34	35	36	37	38	39	40	41
Answer	B	B	B	D	B	A	B	A

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PART 6 of 12

CLASS 10 SCIENCE

CHEMICAL REACTIONS AND EQUATIONS

MCQs 42-50

TOP 100 MCQs WITH ANSWERS

42. Which of the following shows the correct balanced equation for the reaction of methane with oxygen?

A) $CH_4 + O_2 \rightarrow CO_2 + H_2O$
B) $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$
C) $CH_4 + 2O_2 \rightarrow 2CO_2 + 2H_2O$
D) $2CH_4 + O_2 \rightarrow 2CO_2 + H_2O$

Answer: B) $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$

43. Gas X is produced when aluminium reacts with dilute HCl. Gas X is:

A) Oxygen B) Hydrogen
C) Nitrogen D) Carbon dioxide

Answer: B) Hydrogen

44. Which of the following is a redox reaction?

A) $Zn + 2HCl \rightarrow ZnCl_2 + H_2$
B) $CaCO_3 \rightarrow CaO + CO_2$
C) $HCl + NaOH \rightarrow NaCl + H_2O$
D) $2KClO_3 \rightarrow 2KCl + 3O_2$

Answer: A) $Zn + 2HCl \rightarrow ZnCl_2 + H_2$

45. $FeSO_4$ solution turns green when some drops of KOH solution are added. This is due to the formation of:

A) $Fe(OH)_2$ B) $Fe(OH)_3$
C) Fe_2O_3 D) FeO

Answer: A) $Fe(OH)_2$

46. Which of the following is a scenario of oxidation?

A) Addition of oxygen
B) Addition of hydrogen
C) Removal of oxygen
D) Removal of hydrogen

Answer: A) Addition of oxygen

47. When copper is heated in air, it forms a black coating of:

A) CuO B) Cu_2O
C) $CuSO_4$ D) $Cu(OH)_2$

Answer: A) CuO

48. Which of the following is a combination reaction?

A) $2SO_2 + O_2 \rightarrow 2SO_3$
B) $CaCO_3 \rightarrow CaO + CO_2$
C) $Zn + 2HCl \rightarrow ZnCl_2 + H_2$
D) $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$

Answer: A) $2SO_2 + O_2 \rightarrow 2SO_3$

49. Baking soda is sodium hydrogen carbonate. Its formula is:

A) Na_2CO_3 B) $NaHCO_3$
C) $NaOH$ D) Na_2SO_4

Answer: B) $NaHCO_3$

50. A balanced equation has the same number of atoms of each element on:

A) Left side only B) Right side only
C) Both left and right sides
D) Neither side

Answer: C) Both left and right sides

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ANSWER KEY (PART 6)

Q.No.	42	43	44	45	46	47	48	49	50
Answer	B	B	A	A	A	A	A	B	C

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PART 7 of 12

CLASS 10 SCIENCE

CHEMICAL REACTIONS AND EQUATIONS

MCQs 51-58

TOP 100 MCQs WITH ANSWERS

51. Which of the following is a decomposition reaction?

A) $2KClO_3 \rightarrow 2KCl + 3O_2$
B) $Zn + 2HCl \rightarrow ZnCl_2 + H_2$
C) $CaCO_3 \rightarrow CaO + CO_2$
D) $2Na + Cl_2 \rightarrow 2NaCl$

Answer: A) $2KClO_3 \rightarrow 2KCl + 3O_2$

52. In a combination reaction, two or more reactants combine to form:

A) Two products
B) One product
C) No product
D) Many products

Answer: B) One product

53. Which of the following is a double displacement reaction?

A) $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$
B) $2H_2 + O_2 \rightarrow 2H_2O$
C) $2KClO_3 \rightarrow 2KCl + 3O_2$
D) $C + O_2 \rightarrow CO_2$

Answer: A) $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$

54. Which type of reaction is represented by $A + BC \rightarrow AC + B$?

A) Combination reaction
B) Decomposition reaction
C) Single displacement reaction
D) Double displacement reaction

Answer: C) Single displacement reaction

55. Which of the following represents a single displacement reaction?

A) $Na_2SO_4 + BaCl_2 \rightarrow BaSO_4 + 2NaCl$
B) $Zn + 2HCl \rightarrow ZnCl_2 + H_2$
C) $2H_2 + O_2 \rightarrow 2H_2O$
D) $CaCO_3 \rightarrow CaO + CO_2$

Answer: B) $Zn + 2HCl \rightarrow ZnCl_2 + H_2$

56. Which gas is released when dilute HCl is added to a metal carbonate?

A) Oxygen
B) Hydrogen
C) Carbon dioxide
D) Nitrogen

Answer: C) Carbon dioxide

57. Which of the following is an example of a neutralization reaction?

A) $HCl + NaOH \rightarrow NaCl + H_2O$
B) $Zn + 2HCl \rightarrow ZnCl_2 + H_2$
C) $CaCO_3 \rightarrow CaO + CO_2$
D) $2H_2 + O_2 \rightarrow 2H_2O$

Answer: A) $HCl + NaOH \rightarrow NaCl + H_2O$

58. Thermal decomposition of calcium carbonate is an example of:

A) Combination reaction
B) Decomposition reaction
C) Displacement reaction
D) Redox reaction

Answer: B) Decomposition reaction

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ANSWER KEY (PART 7)

Q. No.	51	52	53	54	55	56	57	58
Answer	A	B	A	C	B	C	A	B

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PART 8 of 12

CLASS 10 SCIENCE

CHEMICAL REACTIONS AND EQUATIONS

MCQs 59-66

TOP 100 MCQs WITH ANSWERS

59. Which of the following is a redox reaction?

A) $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$
B) $2H_2 + O_2 \rightarrow 2H_2O$
C) $CaCO_3 \rightarrow CaO + CO_2$
D) $AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$

Answer: A) $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$

60. Which of the following reactions is used in the preparation of oxygen in the laboratory?

A) $2H_2O_2 \rightarrow 2H_2O + O_2$
B) $2KClO_3 \rightarrow 2KCl + 3O_2$
C) $2H_2 + O_2 \rightarrow 2H_2O$
D) $CaCO_3 \rightarrow CaO + CO_2$

Answer: B) $2KClO_3 \rightarrow 2KCl + 3O_2$

61. Which of the following is an example of a single displacement reaction?

A) $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$
B) $2H_2 + O_2 \rightarrow 2H_2O$
C) $CaCO_3 \rightarrow CaO + CO_2$
D) $2Na + Cl_2 \rightarrow 2NaCl$

Answer: A) $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$

62. Which of the following is a precipitate in the reaction between silver nitrate and sodium chloride?

A) $NaNO_3$ B) $AgCl$
C) $NaCl$ D) $AgNO_3$

Answer: B) $AgCl$

63. Which of the following is not a sign of a chemical reaction?

A) Change in colour
B) Change in state
C) Evolution of gas
D) Formation of precipitate

Answer: B) Change in state

64. Why is a balanced chemical equation important?

A) It shows the colour change
B) It shows the temperature change
C) It satisfies the law of conservation of mass
D) It increases the speed of reaction

Answer: C) It satisfies the law of conservation of mass

65. Which of the following is formed when quicklime reacts with water?

A) $CaCO_3$ B) $Ca(OH)_2$
C) CaO D) CO_2

Answer: B) $Ca(OH)_2$

66. Which of the following reactions is represented by the equation $A + BC \rightarrow AC + B$?

A) Combination reaction
B) Decomposition reaction
C) Displacement reaction
D) Double displacement reaction

Answer: C) Displacement reaction

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ANSWER KEY (PART 8)								
Q. No.	59	60	61	62	63	64	65	66
Answer	A	B	A	B	B	C	B	C

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PART 9 of 12

CLASS 10 SCIENCE

CHEMICAL REACTIONS AND EQUATIONS

MCQs 67-74

TOP 100 MCQs WITH ANSWERS

67. Which of the following is an example of a combination reaction?

A) $CaO + H_2O \rightarrow Ca(OH)_2$
B) $Zn + 2HCl \rightarrow ZnCl_2 + H_2$
C) $2AgBr \rightarrow 2Ag + Br_2$
D) $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$

Answer: A) $CaO + H_2O \rightarrow Ca(OH)_2$

68. Which of the following is an example of a decomposition reaction?

A) $2H_2O \rightarrow 2H_2 + O_2$
B) $CaCO_3 \rightarrow CaO + CO_2$
C) $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$
D) $N_2 + 3H_2 \rightarrow 2NH_3$

Answer: A) $2H_2O \rightarrow 2H_2 + O_2$

69. Which of the following is an example of a displacement reaction?

A) $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$
B) $2Na + Cl_2 \rightarrow 2NaCl$
C) $CaO + H_2O \rightarrow Ca(OH)_2$
D) $2KClO_3 \rightarrow 2KCl + 3O_2$

Answer: A) $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$

70. Which of the following is an example of a double displacement reaction?

A) $AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$
B) $2H_2 + O_2 \rightarrow 2H_2O$
C) $CaO + CO_2 \rightarrow CaCO_3$
D) $Zn + 2HCl \rightarrow ZnCl_2 + H_2$

Answer: A) $AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$

71. Which of the following reactions is NOT a redox reaction?

A) $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$
B) $2H_2 + O_2 \rightarrow 2H_2O$
C) $AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$
D) $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$

Answer: C) $AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$

72. In which of the following reactions, hydrogen is oxidized?

A) $2H_2 + O_2 \rightarrow 2H_2O$
B) $Zn + 2HCl \rightarrow ZnCl_2 + H_2$
C) $SO_2 + 2H_2S \rightarrow 3S + 2H_2O$
D) $2Na + 2H_2O \rightarrow 2NaOH + H_2$

Answer: A) $2H_2 + O_2 \rightarrow 2H_2O$

73. Which of the following types of reaction always involves a change in oxidation state?

A) Combination reaction
B) Displacement reaction
C) Redox reaction
D) Neutralization reaction

Answer: C) Redox reaction

74. A reaction in which heat is released to the surroundings is called:

A) Endothermic reaction
B) Exothermic reaction
C) Oxidation reaction
D) Decomposition reaction

Answer: B) Exothermic reaction

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ANSWER KEY (PART 9)								
Q. No.	67	68	69	70	71	72	73	74
Answer	A	A	A	A	C	A	C	B

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CLASS 10 SCIENCE

CHEMICAL REACTIONS AND EQUATIONS

MCQs 75-82

TOP 100 MCQs WITH ANSWERS

75. Which of the following reactions is used in the extraction of metals by reduction process?

A) $2H_2 + O_2 \rightarrow 2H_2O$
B) $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$
C) $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$
D) $CaCO_3 \rightarrow CaO + CO_2$

Answer: C) $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$

76. Which of the following is a neutralization reaction?

A) $HCl + NaOH \rightarrow NaCl + H_2O$
B) $CaCO_3 \rightarrow CaO + CO_2$
C) $2H_2 + O_2 \rightarrow 2H_2O$
D) $Zn + 2HCl \rightarrow ZnCl_2 + H_2$

Answer: A) $HCl + NaOH \rightarrow NaCl + H_2O$

77. Which gas gives a 'pop' sound when a burning splint is brought near it?

A) Oxygen B) Hydrogen
C) Carbon dioxide D) Nitrogen

Answer: B) Hydrogen

78. Which of the following reactions is not a decomposition reaction?

A) $CaCO_3 \rightarrow CaO + CO_2$
B) $2KClO_3 \rightarrow 2KCl + 3O_2$
C) $2H_2O \rightarrow 2H_2 + O_2$
D) $Zn + 2HCl \rightarrow ZnCl_2 + H_2$

Answer: D) $Zn + 2HCl \rightarrow ZnCl_2 + H_2$

79. Which of the following is formed when methane burns in air?

A) CO_2 and H_2O
B) CO and H_2
C) HCl and H_2
D) SO_2 and H_2O

Answer: A) CO_2 and H_2O

80. Which of the following is an example of an oxidation reaction?

A) $2H_2 + O_2 \rightarrow 2H_2O$
B) $CuO + H_2 \rightarrow Cu + H_2O$
C) $2Mg + O_2 \rightarrow 2MgO$
D) $CaCO_3 \rightarrow CaO + CO_2$

Answer: C) $2Mg + O_2 \rightarrow 2MgO$

81. What happens when quicklime is added to water?

A) It absorbs heat
B) It evolves heat
C) It changes into gas
D) It remains unchanged

Answer: B) It evolves heat

82. Which of the following is a precipitation reaction?

A) $AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$
B) $2H_2 + O_2 \rightarrow 2H_2O$
C) $Zn + 2HCl \rightarrow ZnCl_2 + H_2$
D) $CaCO_3 \rightarrow CaO + CO_2$

Answer: A) $AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$

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ANSWER KEY (PART 10)

Q. No.	75	76	77	78	79	80	81	82
Answer	C	A	B	D	A	C	B	A

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PART 11 of 12

CLASS 10 SCIENCE

CHEMICAL REACTIONS AND EQUATIONS

MCQs 83-90

TOP 100 MCQs WITH ANSWERS

83. Which of the following is an exothermic reaction?

A) Photosynthesis
B) Respiration
C) Dissolution of ammonium nitrate in water
D) Electrolysis of water

Answer: B) Respiration

84. Which of the following changes is temporary?

A) Rusting of iron
B) Burning of paper
C) Melting of ice
D) Formation of curd

Answer: C) Melting of ice

85. Which of the following is a redox reaction?

A) $Zn + 2HCl \rightarrow ZnCl_2 + H_2$
B) $CaCO_3 \rightarrow CaO + CO_2$
C) $2KClO_3 \rightarrow 2KCl + 3O_2$
D) $2H_2 + O_2 \rightarrow 2H_2O$

Answer: A) $Zn + 2HCl \rightarrow ZnCl_2 + H_2$

86. Which of the following is produced when iron reacts with steam?

A) Iron(II) oxide and hydrogen
B) Iron(II) oxide and hydrogen
C) Iron(II) oxide and oxygen
D) Iron and hydrogen

Answer: A) Iron(II) oxide and hydrogen

87. Which of the following represents a displacement reaction?

A) $2SO_2 + O_2 \rightarrow 2SO_3$
B) $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$
C) $CaCO_3 \rightarrow CaO + CO_2$
D) $2KClO_3 \rightarrow 2KCl + 3O_2$

Answer: B) $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$

88. What type of reaction occurs when a metal reacts with an acid?

A) Combination reaction
B) Decomposition reaction
C) Displacement reaction
D) Oxidation reaction

Answer: C) Displacement reaction

89. Which of the following forms a white precipitate when solutions of barium chloride and sodium sulphate are mixed?

A) $BaSO_4$ B) $NaCl$
C) $BaCl_2$ D) Na_2SO_4

Answer: A) $BaSO_4$

90. Which of the following reactions is used in the manufacture of baking powder?

A) $2NaHCO_3 \rightarrow Na_2CO_3 + CO_2 + H_2O$
B) $2NaOH + CO_2 \rightarrow Na_2CO_3 + H_2O$
C) $Na_2CO_3 + 2HCl \rightarrow 2NaCl + CO_2 + H_2O$
D) $CaCO_3 \rightarrow CaO + CO_2$

Answer: A) $2NaHCO_3 \rightarrow Na_2CO_3 + CO_2 + H_2O$

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ANSWER KEY (PART 11)

Q. No.	83	84	85	86	87	88	89	90
Answer	B	C	A	A	B	C	A	A

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PART 12 of 12

CLASS 10 SCIENCE

CHEMICAL REACTIONS AND EQUATIONS

MCQs 91-100

TOP 100 MCQs WITH ANSWERS

91. Which of the following is used in the treatment of indigestion?

A) Sodium bicarbonate
B) Calcium carbonate
C) Quicklime
D) Hydrochloric acid

Answer: A) Sodium bicarbonate

92. A gas that turns lime water milky is released when an acid reacts with a carbonate. The gas is:

A) Oxygen
B) Carbon dioxide
C) Hydrogen
D) Nitrogen

Answer: B) Carbon dioxide

93. Which of the following compounds is thermally unstable?

A) Calcium carbonate
B) Sodium chloride
C) Potassium nitrate
D) Copper(II) sulphate

Answer: A) Calcium carbonate

94. Which of the following is an example of a redox reaction?

A) $Zn + 2HCl \rightarrow ZnCl_2 + H_2$
B) $2KClO_3 \rightarrow 2KCl + 3O_2$
C) $CaCO_3 \rightarrow CaO + CO_2$
D) $AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$

Answer: B) $2KClO_3 \rightarrow 2KCl + 3O_2$

95. Which of the following is used to prevent rancidity of food?

A) Sodium benzoate
B) Sodium metabisulphite
C) Baking soda
D) Common salt

Answer: B) Sodium metabisulphite

96. Which of the following reactions occurs in the presence of a catalyst?

A) Decomposition of calcium carbonate
B) Combination of hydrogen and oxygen
C) Decomposition of hydrogen peroxide
D) Reaction of zinc with dilute HCl

Answer: C) Decomposition of hydrogen peroxide

97. Which of the following is the correctly balanced equation?

A) $H_2 + O_2 \rightarrow H_2O$
B) $2H_2 + O_2 \rightarrow 2H_2O$
C) $H_2 + O_2 \rightarrow 2H_2O$
D) $2H_2 + 2O_2 \rightarrow 2H_2O$

Answer: B) $2H_2 + O_2 \rightarrow 2H_2O$

98. Which of the following is an acidic oxide?

A) CO_2 B) CaO
C) Na_2O D) MgO

Answer: A) CO_2

99. Which of the following is NOT a sign of a chemical reaction?

A) Change in colour
B) Formation of precipitate
C) Change in shape
D) Evolution of gas

Answer: C) Change in shape

100. Which of the following is formed when quicklime reacts with water?

A) Calcium carbonate B) Calcium hydroxide
C) Calcium oxide D) Calcium sulphate

Answer: B) Calcium hydroxide

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ANSWER KEY (PART 12)										
Q. No.	91	92	93	94	95	96	97	98	99	100
Answer	A	B	A	B	B	C	B	A	C	B

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